

Solution Requirements

Date	24 June 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users do not require strict login; the system allows public access to the prediction form.	Low
2	Authorization levels	All users have standard access to submit applications and view their own results.	Low
3	External interfaces	The web app must seamlessly interface with the serialized model/loan_model.pkl and model/scaler.pkl files.	High

4	Transactions processing	The Flask /submit endpoint must process POST requests containing applicant parameters safely.	High
5	Reporting	The system must generate and output a clear "Loan Approved  " or "Loan Rejected  " message.	High
6	Business rules, etc.	The prediction must rely on rules established by the best-selected model (e.g., Random Forest or XGBoost) trained on SMOTE-balanced data.	High
7	Compliance to laws or regulations	Financial data inputted by the user must not be permanently stored without consent (basic privacy compliance).	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction inference using the .pkl files must be lightweight and fast.	Page loads and prediction returns in < 2 seconds.
2	Scalability	The Flask app logic and model inference must be decoupled enough to handle traffic.	Supports up to 500 concurrent prediction requests.

3	Security & Data Privacy	Form inputs must be sanitized to prevent basic injection attacks on the server.	Strict input validation on all numeric and categorical fields.
4	Reliability & Availability	The machine learning model must be consistently available without crashing the Flask server.	99.9% uptime per month.
5	Usability & Accessibility	The home.html, predict.html, and submit.html interfaces must be clean and intuitive.	Responsive design compliant with standard web browsers.